



Problem

Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

$n = 3$

$c = [8, 3, 1, 2]$

There are **3** ways to make change for $n = 3$: $\{1, 1, 1\}$, $\{1, 2\}$, and $\{3\}$.

Submissions

Function Description

Complete the getWays function in the editor below.

getWays has the following parameter(s):

- int n : the amount to make change for
- int $c[m]$: the available coin denominations

Leaderboard

Returns

- int: the number of ways to make change

Input Format

The first line contains two space-separated integers n and m , where:

n is the amount to change

m is the number of coin types

The second line contains m space-separated integers that describe the values of each coin type.

Discussions

Constraints

- $1 \leq c[i] \leq 50$
- $1 \leq n \leq 250$
- $1 \leq m \leq 50$
- Each $c[i]$ is guaranteed to be distinct.

Hints

Solve overlapping subproblems using [Dynamic Programming](#) (DP):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the [overlapping subproblems](#). Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for **0** cents? - How many ways can you make change for **> 0** cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case ($n = 0$).
- The answer may be larger than a **32**-bit integer.

Sample Input 0

```
4 3
1 2 3
```

Change Theme

Language Python 3



```
3 import os
4 import random
5 import re
6 import sys
7 # Complete the 'getWays' function below.
8 # The function is expected to return a LONG_IN
9 # The function accepts following parameters:
10 # 1. INTEGER n
11 # 2. LONG_INTEGER_ARRAY c
12 def getWays(n, c):
13     dp=[0]*(n+1)
14     dp[0]=1
15     for coin in c:
16         for tot in range(coin,n+1):
17             dp[tot]+=dp[tot-coin]
18     return dp[n]
19 if __name__ == '__main__':
20     fptr = open(os.environ['OUTPUT_PATH'], 'w')
21     first_multiple_input = input().rstrip().sp
22     n = int(first_multiple_input[0])
23     m = int(first_multiple_input[1])
24     c = list(map(int, input().rstrip().split())
25     ways = getWays(n, c)
26     fptr.write(str(ways) + '\n')
27     fptr.close()
```

Line: 12 Col: 19

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Test case 6

Compiler Message

Success

